

Volume by Cross Sections

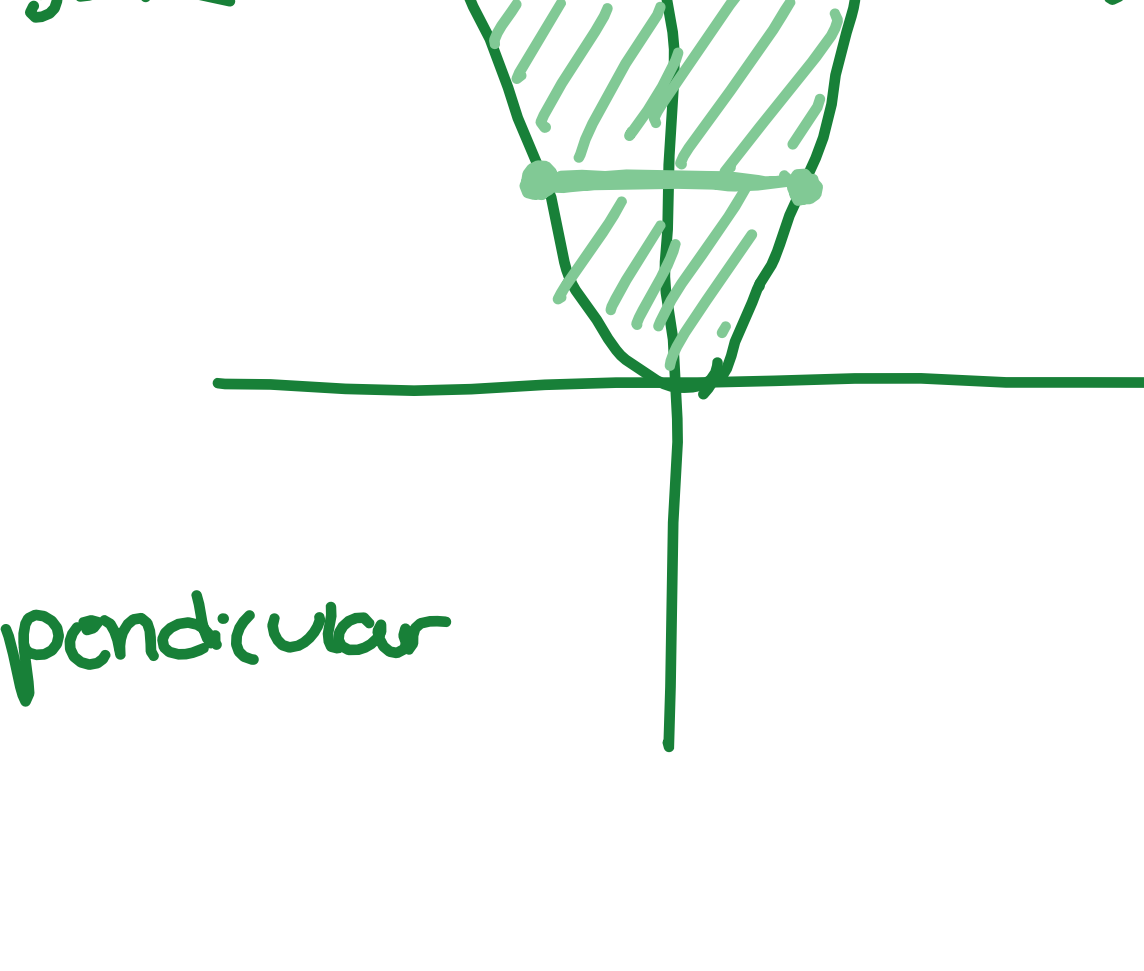
Section 7.2

1-15-26

Ex:

$$y = x^2$$

$$y = 4$$



Square perpendicular

to y-axis

$$A = b^2$$

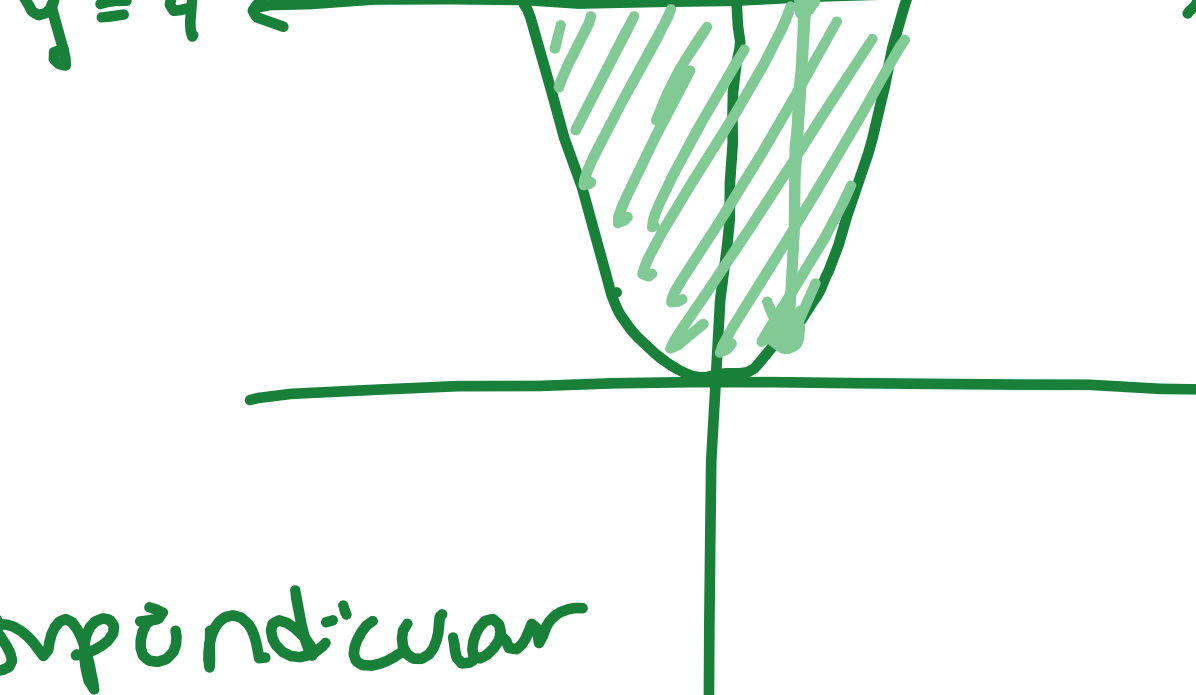
$$A = (\sqrt{y} + \sqrt{y})^2$$

$$\int_0^4 (4y) dy = 2y^2 \Big|_0^4 = \boxed{32}$$

Ex:

$$y = x^2$$

$$y = 4$$



Square perpendicular

to x axis

$$\int_{-2}^2 (4 - x^2)^2 dx = \int_{-2}^2 (16 - 8x + x^4) dx$$

$$= (16x - 4x^2 + \frac{1}{5}x^5) \Big|_{-2}^2$$

$$= \boxed{34.133}$$

Ex:

$$y = x^2$$

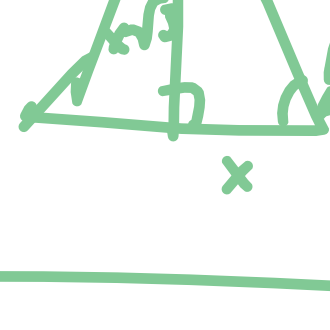
$$y = 4$$



Equilateral Triangles

perpendicular to y-axis

$$A = \frac{1}{2}bh = \frac{\sqrt{3}}{4} b^2$$



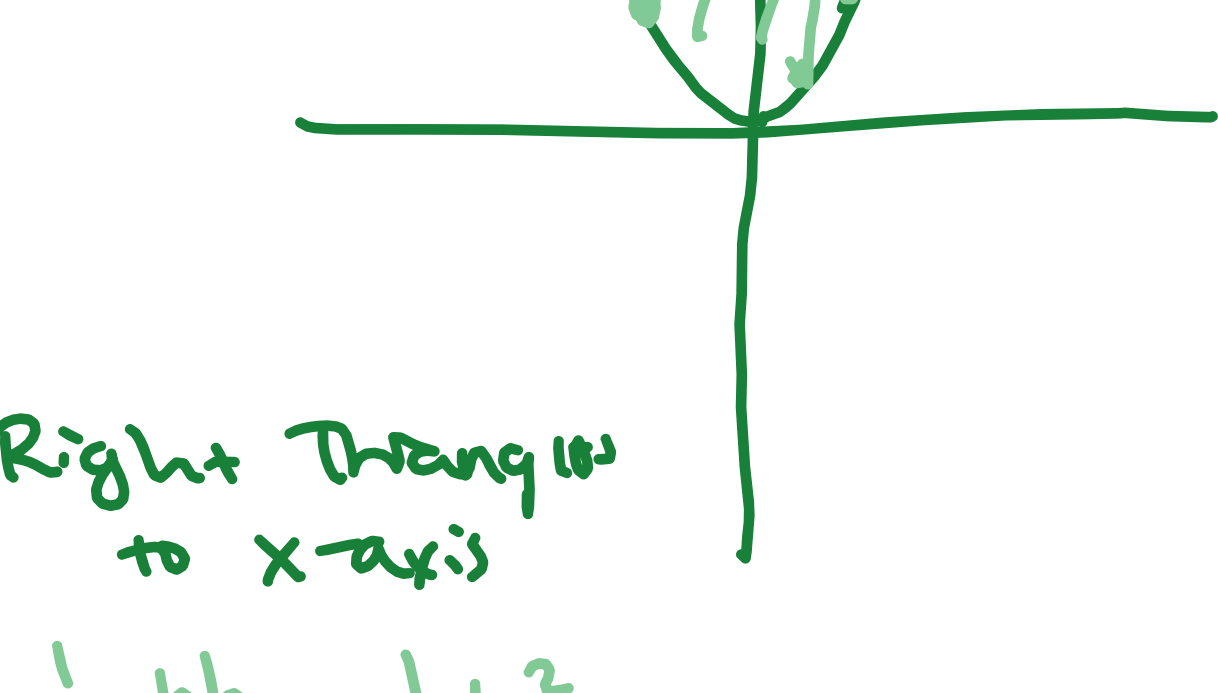
$$\frac{1}{2} \cdot \frac{\sqrt{3}}{2}$$

$$\frac{\sqrt{3}}{4} \int_0^4 (2\sqrt{y})^2 dy = \dots$$

Ex:

$$y = x^2$$

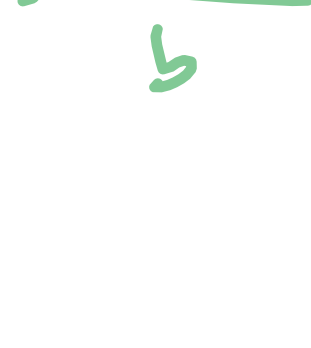
$$y = 4$$



Isosceles Right Triangles

perpendicular to x-axis

$$\frac{1}{2} bh = \frac{1}{2} b^2$$



$$\frac{1}{2} \int_{-2}^2 (4 - x^2) dx$$

$$\text{or}$$

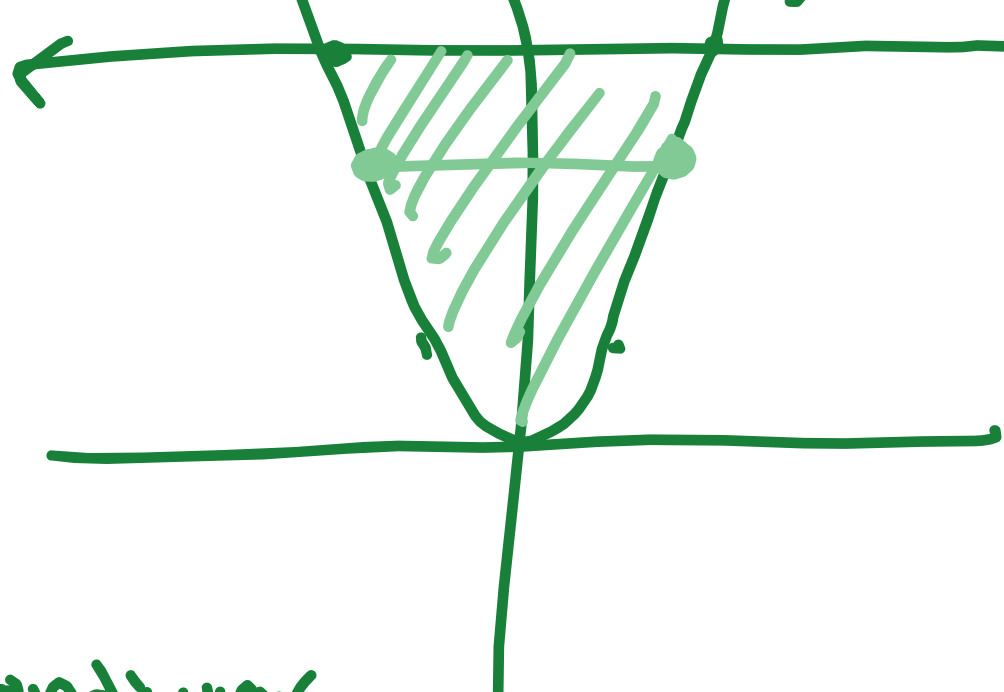
$$\int_0^2 (4 - x^2) dx$$

Ex:

$$y = 4$$

$$y = x^2$$

$$y = 4$$



Rectangle perpendicular

to y-axis, height is 3

$$A = bh \quad A = 3b$$

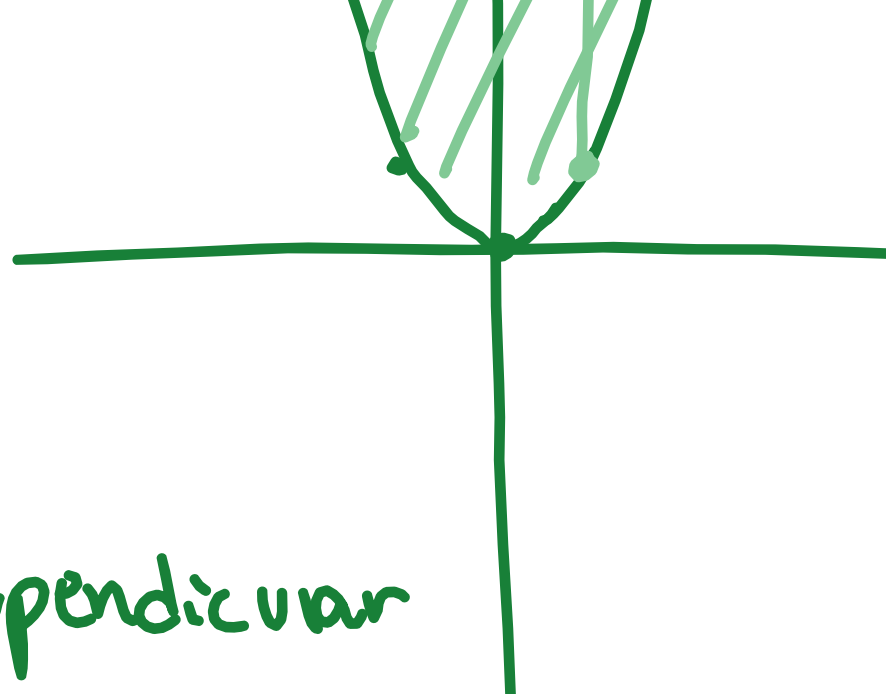
$$3 \int_0^4 (2\sqrt{y}) dy$$

Ex:

$$y = 4$$

$$y = x^2$$

$$y = 4$$



Semi Circle perpendicular

to the x-axis

$$A = \frac{1}{2} \pi r^2$$

$$= \frac{1}{2} \pi \left(\frac{1}{2} B \right)^2$$

$$= \frac{\pi}{8} B^2$$

$$\frac{\pi}{8} \int_{-2}^2 (4 - x^2)^2 dx$$

or

$$\frac{\pi}{4} \int_0^2 (4 - x^2)^2 dx$$